Constitutive theory

From last time...

Using the concrete form for the Lagrangian density, the equations of motion are:

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0,$$

Along with the traction-free boundary conditions:

$$T_{ij} \hat{n}_j = 0.$$

Where we have defined the **first Piola-Kirchhoff** stress by:

$$T_{ij}(\mathbf{x}, t) = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)],$$

Material frame indifference

For a simple hyperelastic material, the general form for the potential energy is:

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, t, \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] \, \mathrm{d}^3\mathbf{x},$$

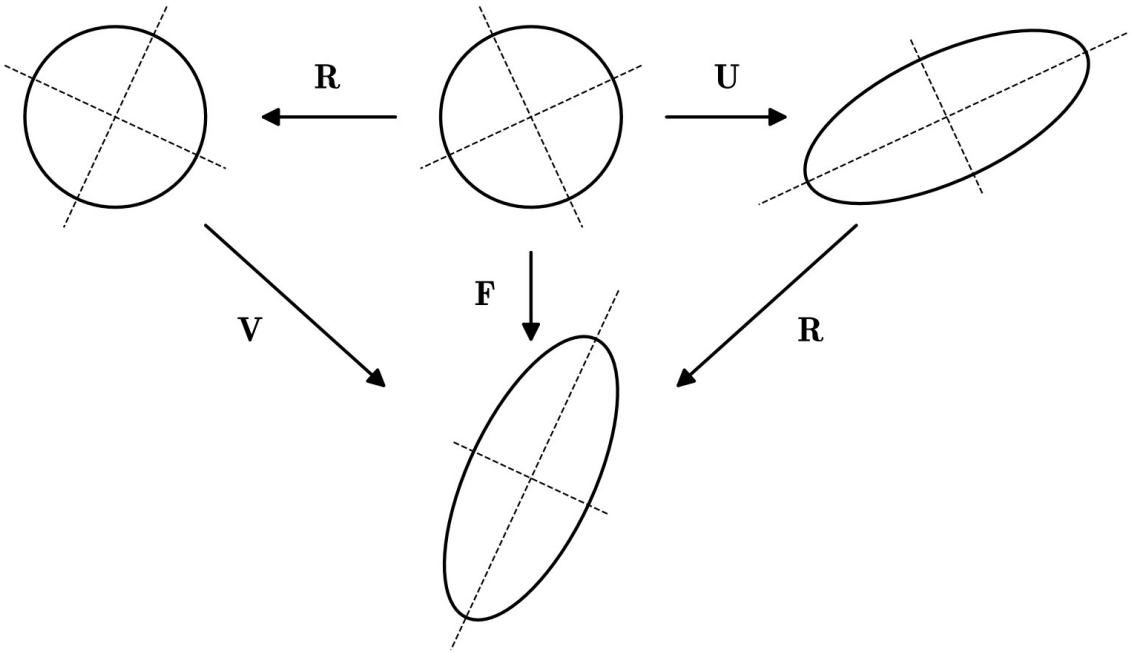
Using invariance under time-translation and superimposed rigid body motions, we showed that:

$$W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)] = W[\mathbf{x}, \mathbf{Q}(t)\mathbf{F}(\mathbf{x}, t)],$$

How do we determine suitable classes of functions for the potential energy density?

The **polar decomposition theorem** shows that

$$\mathbf{F} = \mathbf{RU} = \mathbf{VR},$$



Proof in the handout, but **non-examinable**.

Using the decomposition along with material frame indifference, we obtain:

$$W(\mathbf{x}, \mathbf{F}) = W(\mathbf{x}, \mathbf{RU}) = W(\mathbf{x}, \mathbf{U}),$$

The potential energy density depends only on the stretching part of the deformation gradient.

Due to this result, we can always write:

$$W(\mathbf{x}, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}),$$

Where we have introduced the **right Cauchy-Green deformation tensor**:

$$\mathbf{C} = \mathbf{F}^T\mathbf{F},$$

This is a local measure of change of shape/volume, and from it we define the **finite strain tensor**:

$$\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I})$$

From hereon, we call the potential energy density the **strain energy function** for the material.

From the definition of the first Piola-Kirchhoff stress tensor, we have:

$$T_{ij} = \frac{\partial W}{\partial F_{ij}} = \frac{\partial U}{\partial C_{kl}} \frac{\partial C_{kl}}{\partial F_{ij}}$$

This implies that:

$$T_{ij} = F_{ik} S_{kj},$$

Where we have defined the second Piola-Kirchhoff stress tensor:

$$S_{ij} = 2 \frac{\partial U}{\partial C_{ij}},$$

This is a **symmetric tensor**, with this symmetric being equivalent to the **conservation of angular momentum** – a fact you will work through in the third problem set

Equilibrium and linearisation

At equilibrium, the motion is time-dependent. The equations of motion reduce to:

$$\frac{\partial T_{ij}^0}{\partial x_j} = 0,$$

$$T_{ij}^0 \hat{n}_j = 0.$$

We can *chose* the reference body to be the equilibrium configuration so that the **motion is the identity mapping**.

A solution of the problem is then given by a **stress-free equilibrium**.

We will assume such an equilibrium state for the first part of this course. But in the Earth, due to gravitation and rotation, the equilibrium state is not stress-free.

We consider the disturbance of body about a stress-free equilibrium state:

$$\varphi_i(\mathbf{x}, 0) = x_i + s u_i^0(\mathbf{x}),$$

$$v_i(\mathbf{x}, 0) = s v_i^0(\mathbf{x}).$$

Here *s* is a dimensionless perturbation parameter. We expand the motion in a Taylor series:

$$\varphi_i(\mathbf{x}, t) = x_i + s u_i(\mathbf{x}, t) + O(s^2),$$

Where ***u*** is known as the displacement vector.

All we have to do is put this into the equations of motion, expand to first-order in *s* and see what happens!

In linearising the stress tensor, we find:

$$T_{ij}(\mathbf{x}, t) = s\frac{\partial^2 W}{\partial F_{ij}\partial F_{kl}}(\mathbf{x}, \mathbf{I})\frac{\partial u_k}{\partial x_l} + O(s^2),$$

Motivated by this we introduce the elastic tensor:

$$A_{ijkl}(\mathbf{x}) = \frac{\partial^2 W}{\partial F_{ij}\partial F_{kl}}(\mathbf{x}, \mathbf{I}),$$

Due to the equality of mixed partial derivatives, we note the **hyperelastic symmetry**:

$$A_{ijkl} = A_{klij}.$$

Due to the stress-free equilibrium assumption, you will show that:

$$A_{ijkl} = A_{jikl} = A_{ijlk}$$

Along with the hyperelastic symmetry, this means there are **21 independent elastic moduli**.

The linearised stress depends only on the symmetrised displacement gradient:

$$A_{ijkl}\frac{\partial u_k}{\partial x_l} = A_{ijkl}\frac{1}{2}\left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k}\right),$$

Here we see the **linearised strain tensor**, which is the first-order term in the expansion of the finite strain.

The final linearised equations of motion are:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0,$$

along with boundary conditions:

$$\hat{n}_j A_{ijkl} \frac{\partial u_k}{\partial x_l} = 0.$$

Note that the boundary conditions are applied on the reference boundary without need for further approximation.

We will see later that this is an example of a **wave equation**.

Material symmetries

In general, the following relation, with ***Q*** a rotation matrix, does not hold:

$$W(\mathbf{x}, \mathbf{FQ}) = W(\mathbf{x}, \mathbf{F}),$$

If it does, we say that ***Q*** is in the **material symmetry group** of *W* at the point ***x***.

Material symmetries reflect the invariance of the strain energy with respect to the **orientation of stretching**.

On a crystal scale, they reflect lattice structures.

If the material symmetry group is **all rotation matrices**, the material is **isotropic** at that point.

Recalling frame indifference, this is equivalent to:

$$U(\mathbf{C}) = U(\mathbf{Q}^T\mathbf{C}\mathbf{Q}).$$

But if ***Q*** is an arbitrary rotation matrix, we can select it so that the argument on the RHS is diagonalised.

Thus, in an isotropic material the strain energy depends on ***C*** through its eigenvalues.

From this, and some tedious algebra, we can show that:

$$A_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}),$$

An isotropic material has only **two elastic moduli**!

In an isotropic material with spatially homogeneous properties, the linearised equations about a stress-free equilibrium state reduce to the form:

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} - (\lambda + \mu)\nabla(\nabla \cdot \mathbf{u}) - \mu \nabla^2 \mathbf{u} = \mathbf{0}.$$

This equation is seen in 1B classical dynamics. We have seen where it comes from from first principles, and in this manner we can also see that it is **not suitable** for understanding seismic waves in the Earth.

A material that is not isotropic is said to be **anisotropic**.

A special case are transversely isotropic materials, whose symmetry group comprises rotations about a fixed axis. In this case, it can be shown that the elastic tensor takes the form:

$$A_{ijkl} = (A-2N)\delta_{ij}\delta_{kl} + N(\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk}) + (F-A+2N)(\delta_{ij}\hat{v}_k\hat{v}_l + \hat{v}_i\hat{v}_j\delta_{kl}) \\ + (L-N)(\hat{v}_i\hat{v}_k\delta_{jl} + \hat{v}_j\hat{v}_k\delta_{il} + \hat{v}_j\hat{v}_l\delta_{ik} + \hat{v}_i\hat{v}_l\delta_{jk}) + (A+C-2F-4L)\hat{v}_i\hat{v}_j\hat{v}_k\hat{v}_l,$$

Here there are **five elastic moduli**.

You do not need to learn this expression. And you will always be provided with the necessary form of an elastic tensor within any exam question.

An elastic material is a solid if the symmetry group is a sub-group of rotations.

It is a fluid if it comprises all volume preserving linear transformations.

$$W(\mathbf{x}, \mathbf{F}) = V(\mathbf{x}, J),$$

Physically, a solid's potential energy is affected by changes of shape and volume relative to a reference state, while a fluid cares only about changes in volume.

The elastic tensor then reduces to:

$$A_{ijkl} = \kappa \delta_{ij} \delta_{kl},$$

Within the linearised theory, an elastic fluid is just an elastic solid with **vanishing shear modulus**.